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MS-CyNum
TP Network Access Control (NAC)
AAA, Radius, 802.1x, EAP

1 Introduction

Part of this practical assignment is inspired by the practical work proposed in [1,2,3]. In addition, Serge Bordères' book "Authentification Réseau avec Radius" [4] is an excellent reference for further study of the various concepts covered in this practical assignment.

Without loss of generality, this practical work will be carried out on Linux Ubuntu machines and Cisco routers and switches. Start by logging in as root on the Linux stations (login: root; pwd: linadm).

2 Work architecture and installation of the "freeradius" Radius server

First, set up a local network 192.168.0.0/24 between two stations and a router. The router interface will have the IP address 192.168.0.1, the first station (client) will have 192.168.0.2, and the second station (Radius server) will have 192.168.0.3. On your switch, connect all the cables: the cable from the client to the switch, the cable from the switch to the router, and the cable from the switch to the second station. (client) will be 192.168.0.2 and the second station (Radius server) will be 192.168.0.3. On your switch, connect all cables to ports 1-12 so that all your interfaces are in VLAN 1. Test IP connectivity between the two stations and the router.

Even though we will ultimately use only one machine as the radius server (192.168.0.3), the installation and configuration of the radius server (Sections 2 and 3) can be done on both machines so that all students can learn how to use freeradius.

Start by installing freeradius with the command:

```
sudo apt-get install freeradius
```

Consult the freeradius command help to find the options that allow you to verify that freeradius is properly installed while displaying debug information.

3 Configuring the "freeradius" Radius server

FreeRadius has several configuration files to describe its environment. These files are located in the /etc/freeradius/ directory (or /etc/freeradius/3.0/ depending on your installation). Start by looking at this folder. Consult the dictionary file to familiarize yourself with the concept of attributes.

The three most important files we will study in this practical exercise are: clients.conf, radiusd.conf, and users.

3.1 clients.conf configuration file

This file contains information about clients (NAS) and the secret they must share with the radius server. This file can contain several entries in the following format:

```
client <short-name> {
        <attribute> = <value>
}
```

The most important attributes are "ipaddr" (or "ipv6addr" in IPv6), "secret," "shortname," and "nas-type." The "secret" attribute corresponds to a secret shared between clients and the NAS server and is used to secure communication between the server and its clients (RADIUS message authentication and encryption of some attributes). In versions 1.x, the string after the word client was the client's IP address. In versions 2.x, the IP address is configured via the ipaddr or ipv6addr fields. For compatibility reasons, the 1.x format is still accepted in versions 2.x.

A simple example is provided in the following file. Save the old file in clients.conf.orig and update your clients.conf file.

```
client 192.168.0.0/24 {
        secret = mysecret
}
```

3.2 radiusd.conf configuration file

This configuration file contains the modules, options, and settings for the radius server. In summary, radiusd.conf consists of the following sections:

- Radiusd service parameters: this section contains the declaration of access paths and listening ports, as well as the inclusion of the clients.conf file seen above.
- Module declaration section: this section contains the declaration of the modules required for operation.
- Instantiate section: optional section used to preload some modules.
- **Authorize** section: Contains the modules that must be executed to prepare the intermediate tables required for the authentication phase. These tables contain the values of the attributes sent by the authentication requests on one side and the information retrieved from the authentication databases on the other.
- **Authenticate** section: contains the list of modules that will be executed by the server for authentication. This section can remain empty if it is a simple password verification.
- **PreAcct** section: contains the list of modules called before accounting (traceability/accounting).

— **Accounting** section: contains the list of modules that manage accounting (traceability/accounting)
— Session Section: contains the list of modules that manage simultaneous sessions;
— Post-Auth Section: contains the actions to be launched after authentication;
— Pre-Proxy Section: contains the modules called before proxying a packet
— Post-Proxy section: contains the modules called when the response arrives from the proxy.

At the end of the **"radiusd.conf"** file, find the file that contains the definitions of the "Authorize," "Authenticate," "Accounting," etc. sections.

From now on, we will only work with the **"default"** file without modifying **ra-diusd.conf**. Only four sections will be of interest to us during this practical exercise: **Authorize, Authorize, Preacct, and Accounting**. It should be noted that the two modules we will use during this practical exercise are:

o **preprocess**: this module analyzes the request from the NAS in order to fill in the "request" attributes;

o **files**: this module allows you to use an authentication database in the form of a local file. It consults this local file and fills in the "control" and "response" attributes based on it;

o The **pap**, **chap**, and **mschap** modules manage authentication methods that we will discuss during this lab. When referenced in the "authorize" section, they search for the attributes associated with the authentication method and set the "Auth-Type" attribute. When referenced in the "authenticate" section, they execute the associated authentication method.

o The **eap** module manages the EAP protocol, which is a protocol for transporting authentication messages (see course). When referenced in the "authorize" section, it searches for EAP-Message attributes in the request and sets the "Auth-Type := EAP" attribute. When referenced in the "authenticate" section, it executes the authentication procedure for the method it contains.

Before modifying it, copy it to another file **named "default.orig**." Then replace the contents of the default file with the following minimum configuration:

```
authorize {
preprocess
files
pap
chap
eap
}

authenticate {
Auth-Type PAP {
                pap
                }
Auth-Type CHAP {
                chap
                }
Auth-Type EAP {
                eap
                }
}

preacct {
}
accounting {
}
session {
}
post-auth {
}
pre-proxy {
}
post-proxy {
}
```

3.3 users configuration file

The users file is the local authentication file used by the "files" module for authentication and authorization. This file consists of fields that correspond to conditions. DEFAULT corresponds to any value of "User-Name." The line Fall-Through = 1 allows the analysis of the following fields to continue.

```
DEFAULT
        Tunnel-Type =
        VLAN, Fall-Through
        = 1


DEFAULT Nas-Port == 10, Nas-IP-Address == 192.168.0.2
        Idle-Timeout = 600,
        Fall-Through = 1

walid Cleartext-Password := "test"
        Tunnel-Private-Group-Id = 4

francois Cleartext-Password := "test123"
        Tunnel-Private-Group-Id = 5
```

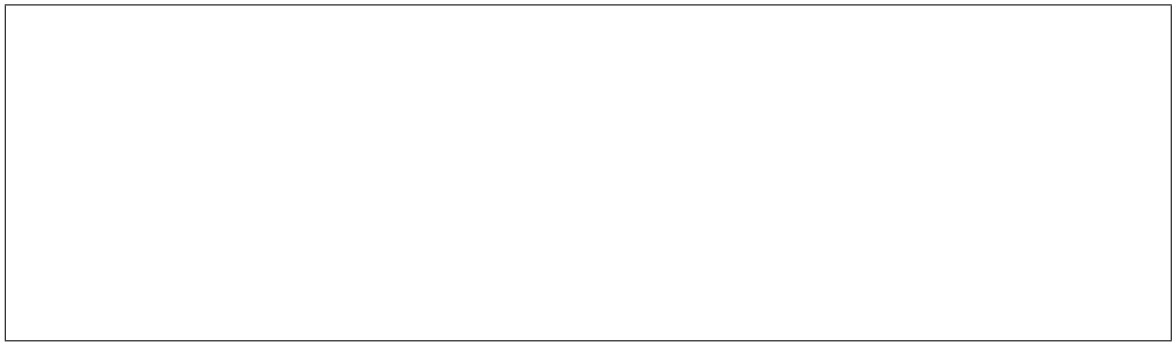
Explain the four sections below. Then save your file in users.orig and replace it with a file similar to the one given in the example above (you can modify and/or add sections).

4 Initial Radius authentication tests (Linux station as NAS client)

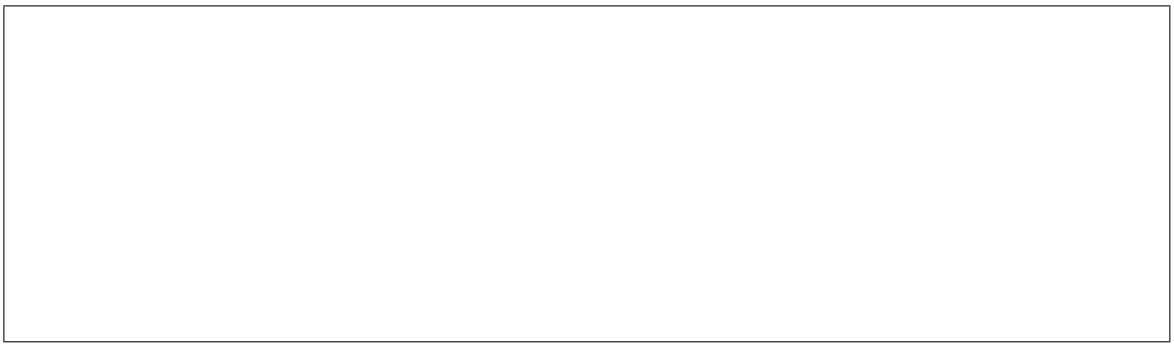
Start this step by launching Wireshark on the client station (192.168.0.2) and on the RADIUS server (192.168.0.3).

Now start the radius server (freeradius command) on the station with the 192.168.0.3 interface. Please note that you must first stop the freeradius service if it is running (sudo service freeradius stop). A more "aggressive" approach would be to stop all processes listening on port 1812, for example (kill -9 $(lsof -t -i :1812)). Also, choose the right option to view the debug information.
On the client side, the Radtest command is a simple command that allows you to test the radius server by querying it. By looking at its manual and the information displayed on the server, build the correct command and test your radius server. Analyze both the displayed results and the exchanged packets captured by Wireshark.

This step is important, so take the time to test valid/invalid secrets, valid/invalid users, valid/invalid passwords, etc. Also change the users file to increase the number of tests.



Test the three authentication methods: PAP, CHAP, and EAP-MD5. Analyze the packets captured each time.



If you would like to learn more about authentication methods with freera-dius, please refer to Chapter 5 of the freeradius guide available online [5].

5 802.1X authentication and dynamic VLAN assignment

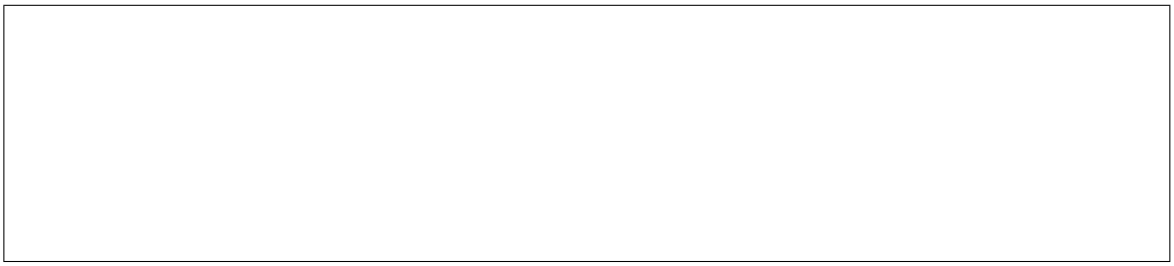
In this section, you will set up 802.1X authentication and associated authorizations (automatic assignments to VLANs). Start by connecting to the switch on your half-bench.

5.1 Configuring the switch as a NAS client and 802.1X controller

On the switch, display the various VLANs that have already been configured (using the "show vlan" command), make a note of them, and note the interfaces to which the client and server stations are connected (you can use the show mac address-table command).

Then configure the IP address of VLAN 1 so that it is in the same IP network as the Radius server. Test the connectivity in your local network.

Now create two other VLANs with IDs 4 and 5 (command "vlan id-vlan").



1.1 Enable the AAA authentication model on the switch.

Switch(config) aaa new-model

1.2 Create a default 802.1X authentication profile.

Switch(config) aaa authentication dot1x default group radius

1.3 Create a default authorization profile.

Switch(config) aaa authorization network default group radius

1.4 Enable 802.1X authentication on the switch.

Switch(config) dot1x system-auth-control

At the interface you want to control (port on the switch to which the client is connected)
2.1 Set the port to Access mode.

Switch(config-int) switchport mode access

2.2 Enable the 802.1X feature on the interface.

Switch(config-int) dot1x pae authenticator

2.3 Condition access to the port on 802.1X authentication.

Switch(config-int) authentication port-control auto

3. Using the radius-server command, configure your Radius server settings (host IP, ports, and shared secret). Here, the ports must be the default ports configured on the radius server: auth-port 1812 and acct-port 1813.

Switch(config) radius-server ...

You can confirm that your Radius client (switch at this level) and the Radius server are working properly with the command.

Switch test aaa group radius <user> <password> port <port-nas> new-code

Disconnect and reconnect the cable from the "Client" station to the switch. Can you ping other stations on the local network? Explain.

5.2 Configuring a Supplicant client and testing authentication

On the "client" station, create a configuration file and fill it with the following clauses:

```
ctrl_interface=/var/run/wpa_supplicant
ctrl_interface_group=0
eapol_version=2
ap_scan=0

network={
        key_mgmt=IEEE8021X
        eap=MD5
        identity="francois"
        password="test123"
        eapol_flags=0
}
```

Now run the wpa-supplicant command and observe the packets exchanged and the messages displayed on the terminals.

wpa_supplicant -c <config-file> -D wired -i ethx

Feel free to unplug/replug the cable and try several users. Analyze the packets exchanged and the messages displayed each time.

5.3 Automatic VLAN assignment (A for Authorization)

On the switch, create the VLANs to which you want to assign users.

On the Radius server, set the following attributes for all your users in the user file (you can use DEFAULT for common attributes):

Tunnel-Type = vlan
Tunnel-Medium-Type = IEEE-802
Tunnel-Private-Group-Id = X

Disconnect the client station cable and reconnect it. Display your VLANs on the switches. Comment.

6 AAA protocol with a router as a NAS client (BONUS)

From now on, we will use the router as a NAS client. Launch Wireshark on the radius server.

Try to connect to the router via telnet from station 192.168.0.2. Does the connection work correctly?

On the router, access the configuration for lines VTY 0 to 4 and enable local authentication mode. Be careful not to select the wrong line type; only VTY lines are used to control Telnet connections.

```
Router(config) line [aux | console | tty | vty] line-number [ending-line-number] Router(config-line) login local
```

You can also create a user

```
username <name> secret <password>
```

Try connecting to the router again via telnet from station 192.168.0.2. Does the connection work correctly? Try switching to enable mode. Set an enable password if necessary (command "enable password" under cisco).

6.1 Router as NAS client: A for Authentication

We will now use an AAA model to ensure authentication using a radius server, as well as rights assignment and traceability.

1. Enable the AAA authentication model on the router.

```
Router(config) aaa new-model
```

2. Use the "aaa authentication" command to create an authentication profile. You will need to find and set the correct authentication method. (Use !).

```
Router(config) aaa authentication login {default | list-name } method1 [ method2... ]
```

3. Access the configuration for VTY lines 0 to 4.

Be very careful not to select the wrong line type; only VTY lines are used to control Telnet connections.

Router(config) line [aux | console | tty | vty] line-number [ending-line-number]

4. Apply the authentication profile created to the VTY lines.

Router(config-line) login authentication {default | list-name}

5. Using the radius-server command, configure your Radius server settings (IP, ports, and shared secret).

Router(config) radius-server

Now test different Telenet connections and analyze the results as well as the messages displayed on the Radius server and the captured messages.

6.2 Router as NAS client: A for Authorization

In this section, you will automatically assign different rights to users following their authentication.

- Start by modifying your users file on the radius server. Cisco uses the "Service-Type" attribute, which can take the values "Administrative-User" and "Login-User." Assign different privileges to your users.
- Define a new authorization scheme. Follow the same steps as for the authentication scheme, but note that "authentication" becomes "authorization" and "exec" replaces "login." Give your authorization list a different name.
- Apply the created schema to the telnet lines.

Now test telnet connections with the different users. Comment on the results.

6.3 Router as NAS client: A for Accounting

Based on the two previous sections, set up an accounting scheme and apply it to the VTY lines.

7 Main references used

1 Educational resources from the University of Pau and Pays d'Adour (AAA and 802.1X practicals).

2 Educational resources from the University of Rennes, Network practicals handout, Adlen Ksentini (802.1X Authentication practical set up by Gilles Guette).

3 Teaching resources from ENSEEIHT Toulouse, The RADIUS protocol, Emmanuel Chapput.

4 Network authentication with Radius: 802.1x - EAP - FreeRadius. Serge Bordères, Eyrolles, 2007.

5 The freeradius implementation guide, Chapter 5 - basic authentication methods, available online at https://networkradius.com/doc/FreeRADIUS-Implementation-Ch5.pdf